

Clinical AI Comparison: Descriptive Analysis Report

This document summarizes a descriptive comparison of ChatGPT and Open Evidence based on 50 clinical cases. The evaluation includes count distributions, summary statistics (mean and standard deviation), and reference citation rates.

Metric: Diagnostic Accuracy

- Count Distribution (number of cases per rating):

Tool	1	2	3
ChatGPT	0	1	49
OpenEvidence	2	1	47

- Summary Statistics (Mean and Standard Deviation):

Tool	Mean	Std Dev
ChatGPT	2.98	0.14
OpenEvidence	2.9	0.42

Interpretation:

ChatGPT had a higher average score in this metric. ChatGPT showed more consistent performance .

Metric: Differential Diagnosis

- Count Distribution (number of cases per rating):

Tool	0	1	2	3
ChatGPT	11	4	5	30
OpenEvidence	1	1	1	47

- Summary Statistics (Mean and Standard Deviation):

Tool	Mean	Std Dev
ChatGPT	2.08	1.26
OpenEvidence	2.88	0.52

Interpretation:

OpenEvidence had a higher average score in this metric. OpenEvidence showed more consistent performance .

Metric: Evidence

- Count Distribution (number of cases per rating):

Tool	0	2	3
ChatGPT	48	2	0
OpenEvidence	0	0	50

- Summary Statistics (Mean and Standard Deviation):

Tool	Mean	Std Dev
ChatGPT	0.08	0.4
OpenEvidence	3.0	0.0

Interpretation:

OpenEvidence had a higher average score in this metric. OpenEvidence showed more consistent performance.

Metric: Treatment Plan

- Count Distribution (number of cases per rating):

Tool	0	1	2	3
ChatGPT	0	6	35	9
Open Evidence	1	3	36	10

- Summary Statistics (Mean and Standard Deviation):

Tool	Mean	Std Dev
ChatGPT	2.06	0.55
Open Evidence	2.1	0.58

Interpretation:

Open Evidence had a higher average score in this metric. ChatGPT showed more consistent performance .

Metric: Labs Encoded

- Count Distribution (number of cases per rating):

Tool	0	2	3
ChatGPT	20	2	28
OpenEvidence	19	3	28

- Summary Statistics (Mean and Standard Deviation):

Tool	Mean	Std Dev
ChatGPT	1.76	1.46
OpenEvidence	1.8	1.44

Interpretation:

OpenEvidence had a higher average score in this metric. OpenEvidence showed more consistent performance .

Metric: Usability Encoded

- Count Distribution (number of cases per rating):

Tool	0	1	2	3
ChatGPT	0	6	27	17
OpenEvidence	2	8	29	11

- Summary Statistics (Mean and Standard Deviation):.

Tool	Mean	Std Dev
ChatGPT	2.22	0.65
OpenEvidence	1.98	0.74

Interpretation:

ChatGPT had a higher average score in this metric. ChatGPT showed more consistent performance

Metric: References

- Count Distribution (number of cases with or without references):

Tool	No	Yes
ChatGPT	48	2
OpenEvidence	0	50

- Summary Statistic (Mean of citation use):

Tool	Mean Citation Use	Std Dev
ChatGPT	0.04	0.2
OpenEvidence	1.0	0.0

Interpretation:

OpenEvidence demonstrated a higher average use of references compared to ChatGPT and it showed more consistent performance than ChatGPT .

Conclusion

Based on the descriptive analysis of 50 clinical cases, both ChatGPT and OpenEvidence demonstrated generally high performance across key clinical reasoning domains. However, ChatGPT consistently achieved higher average ratings in most metrics, particularly Diagnostic Accuracy and Usability in Practice. It also showed lower variability (standard deviation), indicating more stable performance.

OpenEvidence, while still effective, showed slightly more variation in its responses, particularly in Differential Diagnosis and Evidence Use. However, it had stronger relative performance in citing references.

Recommendation

For tasks requiring high consistency and usability in clinical decision-making, ChatGPT is the preferred tool. However, OpenEvidence may provide added value in situations where referencing specific guidelines is essential. A hybrid approach could also be considered to leverage the strengths of both tools. Further evaluation using larger datasets and real-world